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Sexually Dimorphic Neurons in Hypothalamic Neuroendocrine Reproductive Circuits

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Introduction

The hypothalamic-pituitary-gonadal axis contains sexually dimorphic neural circuits that control reproductive functions through specialized neuronal populations expressing distinct molecular markers and steroid hormone receptors. These circuits show sex-specific anatomy, connectivity patterns, and responses to hormonal signals that coordinate reproductive behaviors and neuroendocrine regulation. (5 sources)

The mammalian reproductive neuroendocrine system is fundamentally organized around sexually dimorphic neural circuits within the hypothalamus that coordinate reproductive physiology and behavior through the hypothalamic-pituitary-gonadal axis (Choe et al., 2021). The hypothalamus serves as a critical integration center that coordinates reproductive behaviors and physiological responses with environmental cues through its close anatomical and physiological relationship with the pituitary gland, providing an effective means for neuroendocrine regulation of hormone secretion (Abrahao et al., 2012).

Several hypothalamic nuclei exhibit pronounced sexual dimorphism in both mammals and birds, including the preoptic area (POA), paraventricular nucleus (PVN), and ventromedial nucleus (VMN), along with associated structures like the sexually dimorphic bed nucleus of the stria terminalis and medial amygdala (Choe et al., 2021) (Rosinger et al., 2019). A particularly well-characterized example is the kisspeptin system, which shows notable sexual dimorphism and is critical for brain gender differentiation through regulation of postnatal testosterone secretion (Nejad et al., 2017). In rodents, kisspeptin neurons are concentrated in two major hypothalamic regions: a rostral population in the anteroventral periventricular nucleus (AVPV) within the POA, and a caudal population in the arcuate nucleus (Nejad et al., 2017) (Adachi et al., 2007).

The AVPV represents a classic example of sexual dimorphism, containing a greater number of kisspeptin neurons in females compared to males, a difference established during the perinatal critical period through exposure to testosterone or its metabolites (Nejad et al., 2017) (Adachi et al., 2007). These sexually dimorphic hypothalamic populations express sex steroid receptors and are regulated by gonadal steroids, allowing them to mediate hormonal feedback effects on reproductive function (Nejad et al., 2017). Additional sexually dimorphic neuronal populations include tyrosine hydroxylase-expressing neurons, which are more numerous in female than male mice and appear to regulate sexually dimorphic parental behaviors (Abrahao et al., 2012).

Evidence

(Choe et al., 2021)

The role of sex chromosomes and sex hormones in vocal learning systems.

"One such sexually dimorphic neural circuit is within the hypothalamus, a structure shared across vertebrates (mammals, birds, reptiles, fishes). The hypothalamus, via the hypothalamic-pituitary-adrenal/ gonadal/thyroid axis (HPA/HPG/HPT-axis), controls several reproductive functions, including male mating behavior, female ovulation, and pregnancy in viviparous species (McCarthy et al., 2015)(Nishida et al., 2005)(Sower et al., 2009)(Tripp et al., 2020)Turano et al., 2019;(Wright et al., 2010). Some of the sexually dimorphic hypothalamic nuclei in both mammals and birds include the preoptic area (POA) (Gorski et al., 1980)(Viglietti-Panzica et al., 1986), the paraventricular nucleus (PVN) (Jurkevich et al., 1996)(Rosinger et al., 2019), and the ventromedial nucleus (VMN) (Balthazart, 2002)Chester-Jones et al., 2013;(Matsumoto et al., 1986)Turano et al., 2019;(Wu et al., 2009). Areas influenced by the hypothalamus include the sexually dimorphic bed nucleus of the stria terminalis (BnST) (Jurkevich et al., 1999)Wu et al., 2009), pituitary gland (Ho et al., 2020;MacManes et al., 2017;Nishida et al., 2005), and medial amygdala (MeA)/nucleus Taeniae"

(Abrahao et al., 2012)

Abstracts

"The hypothalamus plays a critical role in coordinating expression of reproductive behaviors and physiological responses with environmental cues. Its close anatomical and physiological relationship with the pituitary gland provides an effective means for coordinating diverse homeostatic processes through neuroendocrine regulation of hormone secretion. The hypothalamus contains sexual dimorphic areas which are different in morphology, density, gene expression and neuronal projections. One of the sexually dimorphic neuronal populations in the hypothalamus is tyrosine hydroxylase expressing (TH-ir) neurons whose number is greater in female than in male mice. The role of the sexual dimorphism of these TH-ir neurons is still unknown"

"Our results indicate that TH-ir lesioned females exhibit lower maternal behavior. This suggests that the sexually dimorphic TH-ir neuronal population in adult mice is involved in regulation of sexually dimorphic parental behavior."

(Rosinger et al., 2019)

A sexually dimorphic distribution of corticotropin-releasing factor receptor 1 in the paraventricular hypothalamus.

"Sex differences in neural structures are generally believed to underlie sex differences reported in anxiety, depression, and the hypothalamic-pituitary-adrenal axis, although the specific circuitry involved is largely unclear. Using a corticotropin-releasing factor receptor 1 (CRFR1) reporter mouse line, we report a sexually dimorphic distribution of CRFR1 expressing cells within the paraventricular hypothalamus (PVN; males > females). Relative to adult levels, PVN CRFR1-expressing cells are sparse and not sexually dimorphic at postnatal days 0, 4, or 21. This suggests that PVN cells might recruit CRFR1 during puberty or early adulthood in a sex-specific manner. The adult sex difference in PVN CRFR1 persists in old mice (20-24 months)Adult gonadectomy (6 weeks)resulted in a significant decrease in CRFR1-immunoreactive cells in the male but not female PVN. CRFR1 cells show moderate co-expression with estrogen receptor alpha (ERα) and high co-expression with androgen receptor, indicating potential mechanisms through which circulating gonadal hormones might regulate CRFR1 expression and function. Finally, we demonstrate that a psychological stressor, restraint stress, induces a sexually dimorphic pattern of neural activation in PVN CRFR1 cells (males >females) as assessed by co-localization with the transcription/neural activation marker phosphorylated CREB. Given the known role of CRFR1 in regulating stress-associated behaviors and hormonal responses, this CRFR1 PVN sex difference might contribute to sex differences in these functions."

(Nejad et al., 2017)

The Role of Kisspeptin in Female Reproduction

"In mammals the anatomy and physiology of the neuroendocrine reproductive axis differs between the genders"

"The Kisspeptin system is apparently critical for brain gender differentiation, acting through the regulation of postnatal T secretion. Distribution of Kisspeptin neurons in the hypothalamus varies between species. In mammals there are 2 major regions of these neurons; a rostral one in the Pre-Optic Area (POA) and a caudal one in the arcuate nucleus, with proportionally more Kisspeptin neurons in the ARC than in the POA region (4,30). In rodents, the POA regions are concentrated in the Anteroventral Periventricular Nucleus (AVPV). Anatomical differences between genders have been reported in the hypothalamus of some species, e.g. the rat AVPV is sexually dimorphic, with a greater number of KISS1 neurons in females compared to males (30). Like most sex differences in the brain, this sexual dimorphism is likely caused during the perinatal critical period by exposure to testosterone (or its metabolites) (31)"

"KISS1 neurons express sex steroid receptors and are regulated by gonadal sex steroids, mediating the effects of estrogen on GnRH neurons"

"In female rodents, the AVPV mediates E2 positive feedback, causing occurrence of pre-ovulatory GnRH/LH surge. Circadian cues also impinge upon AVPA, whereby KISS1 neurons in the AVPV receiving hormonal and temporal signals give rise to timely LH surge (30). In contrast to AVPV, the arcuate nucleus shows no differences between males and females"

(Adachi et al., 2007)

Involvement of anteroventral periventricular metastin/kisspeptin neurons in estrogen positive feedback action on luteinizing hormone release in female rats.

> *"Metastin/kisspeptin, the KiSS-1 gene product, has been identified as an endogenous ligand of GPR54 that reportedly regulates GnRH/LH surges and estrous cyclicity in female rats. The aim of the present study was to determine if metastin/kisspeptin neurons are a target of estrogen positive feedback to induce GnRH/LH surges. We demonstrated that preoptic area (POA) infusion of the anti-rat metastin/kisspeptin monoclonal antibody blocked the estrogen-induced LH surge, indicating that endogenous metastin/kisspeptin released around the POA mediates the estrogen positive feedback effect on GnRH/LH release. Metastin/kisspeptin neurons in the anteroventral periventricular nucleus (AVPV) may be responsible for mediating the feedback effect because the percentage of c-Fos-expressing KiSS-1 mRNA-positive cells to total KiSS-1 mRNA-positive cells was significantly higher in the afternoon than in the morning in the anteroventral periventricular nucleus (AVPV) of high estradiol (E(2))-treated females. The percentage of c-Fos-expressing metastin/kisspeptin neurons was not different between the afternoon and morning in the arcuate nucleus (ARC). Most of the KiSS-1 mRNA expressing cells contain ERalpha immunoreactivity in the AVPV and ARC. In addition, AVPV KiSS-1 mRNA expressions were highest in the proestrous afternoon and lowest in the diestrus 1 in females and were increased by estrogen treatment in ovariectomized animals. On the other hand, the ARC KiSS-1 mRNA expressions were highest at diestrus 2 and lowest at proestrous afternoon and were increased by ovariectomy and decreased by high estrogen treatment. Males lacking the surge mode of GnRH/LH release showed no obvious cluster of metastin/kisspeptin-immunoreactive neurons in the AVPV when compared with high E(2)-treated females, which showed a much greater density of these neurons. Taken together, the present study demonstrates that the AVPV metastin/kisspeptin neurons are a target of estrogen positive feedback to induce GnRH/LH surges in female rats."*

Sexually Dimorphic Brain Regions and Structures

Multiple hypothalamic and limbic brain regions show pronounced sexual dimorphism in cell number, volume, and connectivity patterns, with some regions larger in males (like the sexually dimorphic nucleus of the preoptic area) and others larger in females (like the anteroventral periventricular nucleus). These differences extend beyond reproductive regions to include areas involved in stress, pain processing, and social behaviors. (9 sources)

The sexually dimorphic organization of the mammalian brain extends across multiple interconnected regions that form robust social networks controlling reproductive and social behaviors. The two best known dimorphic brain structures are the sexually dimorphic nucleus of the medial preoptic hypothalamic area (SDN-POA) in rodents, which corresponds to the interstitial nucleus of the anterior hypothalamus (INAH) in humans, and the anteroventral periventricular (AVPV) nucleus (Negri-Cesi, 2015). The first controls male sex behavior and is larger in males than in females, while the second is critical for the cyclic control of ovulation and is larger in females than in males (Negri-Cesi, 2015).

These dimorphic brain regions, including the AVPV, medial preoptic nucleus (MPOA), ventromedial hypothalamus (VMH), bed nucleus of the stria terminalis (BNST), and medial amygdala (MeA), send and receive multiple projections from one another to form robust social networks that communicate with brain reward regions and play an essential role in regulating sexually dimorphic reproductive behaviors (Zilkha et al., 2021). The AVPV is notable as an exception to the typical pattern, being larger in volume, containing more cells, and sending more projections to multiple reproduction-related brain regions in females compared to males (Zilkha et al., 2021).

The medial preoptic area represents a large hypothalamic structure that sends projections to multiple downstream brain regions and is both larger and contains more neurons in males than in females (Zilkha et al., 2021). This region houses various heterogeneous, molecularly defined neuronal clusters, including many sexually dimorphic populations such as androgen receptor (AR)-expressing and estrogen receptor alpha (ESR1)-expressing populations (Zilkha et al., 2021). The sexually dimorphic nucleus can be delineated histologically using conventional Nissl staining or immunohistochemically using calbindin D28K immunoreactivity (He et al., 2013).

Additional sexually dimorphic regions have been identified through systematic analysis, including two adjacent hypothalamic regions: one overlapping with the anterior hypothalamic nucleus (AHN) and another primarily overlapping with the ventral premammillary nucleus (PMv), both of which show specific neuronal subclass differences that provide a cellular basis for sex-specific behaviors like aggression and parental care (Hemminger et al., 2024). The bed nucleus of the stria terminalis shows sexual dimorphism in multiple

subdivisions, with females having larger corticotropin-releasing factor (CRF) neurons in the anterolateral BNST than males, and males exhibiting denser projections from the amygdala to the lateral posterior BNST (Kanaya et al., 2025) (Ide et al., 2013) (Uchida et al., 2019).

Importantly, sexual dimorphism extends beyond hypothalamic reproductive centers to include regions not directly involved with reproductive behaviors, such as the hippocampus, locus coeruleus, amygdala, and other limbic regions (Ponzi et al., 2020). The formation of these dimorphic structures involves neuroestrogens inducing sex differences in glial and immune cells, which play important roles in sexually dimorphic formation of dendritic synapse patterning and cell number differences in specific neuronal groups (Tsukahara et al., 2020).

Evidence

(Negri-Cesi, 2015)

Bisphenol A Interaction With Brain Development and Functions

> *"The 2 best known dimorphic brain structures are the sexual dimorphic nucleus of the medial preoptic hypothalamic area (SDN-POA) in rodents, which correspond to the interstitial nucleus of the anterior hypothalamus (INAH) in humans, and the anteroventral periventricular (AVPV) nucleus. The first one controls male sex behavior and is larger in males than in females; the second one is critical for the cyclic control of ovulation and is larger in females than in males."*

(Zilkha et al., 2021)

The social network: Neural control of sex differences in reproductive behaviors, motivation, and response to social isolation

> *"A few brain regions, most of them located within the hypothalamus, have been identified as critical nodes in the rodent social network and have also been shown to be sexually dimorphic. Within each dimorphic brain region, only specific, molecularly distinct subsets of neurons have been found to be sexually dimorphic and required for one or a few specific displays of social behavior [27]. These brain regions, including the anteroventral periventricular nucleus (AVPV), medial preoptic nucleus (MPOA), ventromedial hypothalamus (VMH), bed nucleus of the stria terminalis (BNST), and medial amygdala (MeA), send and receive multiple projections from one another to form robust 'social networks.' These social networks communicate with brain reward regions and play an essential role in the regulation of most, if not all, sexually dimorphic reproductive behaviors [69,70]"*

> *"A notable exception is the AVPV of the hypothalamus, which is larger in volume, contains more cells, and sends more projections to multiple reproduction-related brain regions in females compared to males [25,34,71,72,76e[79]. Importantly, it also expresses several sexually dimorphic molecularly defined neuronal populations, including the tyrosine hydroxylase (TH)-expressing population, which contains 3e4 times more neurons in females than in males [34,72]"*

> *"A large hypothalamic structure, the MPOA sends projections to multiple downstream brain regions and is both larger and contains more neurons in males than in females [35]. Notably, the MPOA is home to various heterogeneous, molecularly defined, neuronal clusters, including many sexually dimorphic populations, such as androgen receptor (AR)-expressing population and estrogen receptor alpha (ESR1)expressing population [80]"*

> *"At least two different subpopulations within the MPOA were shown to be required for the regulation of pupdirected behavior. The first is the ESR1 þ population, which is highly sexually dimorphic in its distribution and projection patterns [85]"*

> *"The second subpopulation is the galaninexpressing (Gal þ) neurons"*

> *"Another molecularly defined sexually dimorphic VMHvl subpopulation that controls sex-typical behaviors in both sexes is the progesterone receptor (PR)-expressing neurons. This subpopulation is required for the normal display of mating in both sexes and for fighting in males [76]."*

(He et al., 2013)

Development of the sexually dimorphic nucleus of the preoptic area and the influence of estrogen-like compounds

> *"One of the well-defined sexually dimorphic structures in the brain is the sexually dimorphic nucleus, a cluster of cells located in the preoptic area of the hypothalamus. The rodent sexually dimorphic nucleus of the preoptic area can be delineated histologically using conventional Nissl staining or immunohistochemically using calbindin D28K immunoreactivity"*

> *"The sexually dimorphic nucleus has been specifically defined in the brains of human and other mammalian and non-mammalian and includes the third interstitial nucleus of the anterior hypothalamus in humans (Allen et al., 1989)(Allen et al., 1990) , the ovine sexually dimorphic nucleus in the medial preoptic area (Roselli et al., 2004) , the medial preoptic and anterior hypothalamic regions in rhesus monkeys (Byne, 1998) , a specific area in the medial preoptic nucleus in quail (Viglietti-Panzica et al., 1986) , and the*

sexually dimorphic nucleus of the preoptic area in rats (Gorski et al., 1978)(Gorski et al., 1980) . The human sexually dimorphic nucleus of the preoptic area is located in the medial part of the preoptic area, between the dorsolateral supraoptic nucleus and the rostral pole of the paraventricular nucleus (Hofman et al., 1989)"

"As demonstrated in laboratory animal studies, the sexually dimorphic nucleus is critically implicated in sexual behavior (Arendash et al., 1983)(Rayen et al., 2013)"

"the sexually dimorphic nucleus of the preoptic area are positive for tyrosine hydroxylase (TH), yet many TH-positive axonand synapse-like structures are found there"

"The sexually dimorphic nucleus of the preoptic area is highlighted by calbindin-D28K immunoreactivity: no TH-positive cells were found, but fine axon-like projections/synaptic structures were seen"

"the anteroventral periventricular nucleus of the hypothalamus (AVPV) consists mainly of TH-positive neurons"

"Neural stem cell activity has been identified in the hypothalamus (Dahiya et al., 2011)(Ahmed et al., 2008)(Kokoeva et al., 2005) (Kokoeva et al., 2007)(Pérez-Martín et al., 2010) , a territory which includes the sexually dimorphic nucleus of the preoptic area. Neural progenitor cells from the ependymal layer of the 3 rd ventricle migrate and differentiate into neurons in the hypothalamus (Ernst et al., 2005)(Xu et al., 2005)(Park et al., 1996)"

(Hemminger et al., 2024)

Spatial Single-Cell Mapping of Transcriptional Differences Across Genetic Backgrounds in Mouse Brains

"We identified two adjacent regions within the hypothalamus that exhibited significant sexual dimorphism. Region 1 overlaps with the anterior hypothalamic nucleus (AHN), while Region 2 primarily overlaps with the ventral premammillary nucleus (PMv). Both regions are known to be sexually dimorphic (Vries et al., 2002) , though they have been studied to different extents"

"Our findings of specific neuronal subclass differences help further refine the molecular resolution of its dimorphic nature and providing a cellular basis for its role in sex-specific behaviors like aggression and parental care. Region 2, the PMv, is a well-established sexually dimorphic region, with roles in maternal aggression 38 , reproductive control 39 , male social behavior 40 , and intermale aggression 41 . Previous studies identified sexually dimorphic neuron populations within the PMv (e.g., PMv-DAT 42 , PMv-PACAP 43), complementing our systematic identification of subclass-level abundance changes."

(Kanaya et al., 2025)

Neonatal testosterone exposure alleviates female-specific severity of formalin-induced inflammatory pain in mice

"Brain regions such as the anteroventral periventricular nucleus of the hypothalamus and the principal nucleus of the bed nucleus of the stria terminalis (BNST) are sexually dimorphic nuclei related to reproductive functions. Both nuclei differ in cell number and areal size between the sexes (Hines et al., 1992)(Kanaya et al., 2014)(Sumida et al., 1993). Regarding the lateral posterior BNST, males exhibit denser projections from the amygdala to this nucleus, a pattern observed in both mice and humans"

"The BNST modulates pain sensitivity by releasing corticotropin-releasing factor (CRF) from neurons in the anterolateral subdivision (Ide et al., 2013). Female mice have larger CRF neurons in the anterolateral BNST than male mice (Uchida et al., 2019). Dopaminergic projection from the periaqueductal gray (PAG) to the BNST, which preferentially targets the dorsal part, including the anterolateral subdivision (Gungor et al., 2016), drives pain-related behaviors differently between male and female mice (Yu et al., 2021b)."

(Ide et al., 2013)

Opposing Roles of Corticotropin-Releasing Factor and Neuropeptide Y within the Dorsolateral Bed Nucleus of the Stria Terminalis in the Negative Affective Component of Pain in Rats

"Pain is a complex experience composed of sensory and affective components. Although the neural systems of the sensory component of pain have been studied extensively, those of its affective component remain to be determined. In the present study, we examined the effects of corticotropin-releasing factor (CRF) and neuropeptide Y (NPY) injected into the dorsolateral bed nucleus of the stria terminalis (dlBNST) on pain-induced aversion and nociceptive behaviors in rats to examine the roles of these peptides in affective and sensory components of pain, respectively. In vivo microdialysis showed that formalin-evoked pain enhanced the release of CRF in this brain region. Using a conditioned place aversion (CPA) test, we found that intra-dlBNST injection of a CRF1 or CRF2 receptor antagonist suppressed pain-induced aversion. Intra-dlBNST CRF injection induced CPA even in the absence of pain stimulation. On the other hand, intra-dlBNST NPY injection suppressed pain-induced aversion. Coadministration of NPY inhibited CRF-induced CPA. This inhibitory effect of NPY was blocked by coadministration of a Y1 or Y5 receptor antagonist. Furthermore, whole-cell patch-clamp electrophysiology in dlBNST slices revealed that CRF increased neuronal excitability specifically in type II dlBNST neurons, whereas NPY decreased it in these neurons. Excitatory effects of CRF on type II dlBNST neurons were suppressed by NPY. These results have uncovered some of the neuronal mechanisms underlying the affective component of pain by showing opposing roles of intra-dlBNST CRF and NPY in pain-induced aversion and opposing actions of these peptides on neuronal excitability converging on the same target, type II neurons, within the dlBNST."

(Uchida et al., 2019)

Female-biased sexual dimorphism of corticotropin-releasing factor neurons in the bed nucleus of the stria terminalis

> *"The bed nucleus of the stria terminalis (BNST) contains the highest density of corticotropin-releasing factor (CRF)-producing neurons in the brain. CRF-immunoreactive neurons show a female-biased sexual dimorphism in the dorsolateral BNST in the rat. Since CRF neurons cannot be immunostained clearly with available CRF antibodies in the mouse, we used a mouse line, in which modified yellow fluorescent protein (Venus) was inserted to the CRF gene, and the Neo cassette was removed, to examine the morphological characteristics of CRF neurons in the dorsolateral BNST. Developmental changes of CRF neurons were examined from postnatal stages to adulthood. Gonadectomy (GDX) was carried out in adult male and female mice to examine the effects of sex steroids on the number of CRF neurons in the dorsolateral BNST. The number of Venus-expressing neurons, stained by immunofluorescence, was compared between male and female mice over the course of development. GDX was carried out in adult mice. Immunohistochemistry, in combination with Nissl staining, was carried out, and the effects of sex or gonadal steroids were examined by estimating the number of Venus-expressing neurons, as well as the total number of neurons or glial cells, in each BNST subnucleus, using a stereological method. Most Venus-expressing neurons co-expressed Crf mRNA in the dorsolateral BNST. They constitute a group of neurons without calbindin immunoreactivity, which makes a contrast to the principal nucleus of the BNST that is characterized by calbindin immunostaining. In the dorsolateral BNST, the number of Venus-expressing neurons increased across developmental stages until adulthood. Sexual difference in the number of Venus-expressing neurons was not evident by postnatal day 5. In adulthood, however, there was a significant female predominance in the number of Venus expressing neurons in two subnuclei of the dorsolateral BNST, i.e., the oval nucleus of the BNST (ovBNST) and the anterolateral BNST (alBNST). The number of Venus-expressing neurons was smaller significantly in ovariectomized females compared with proestrous females in either ovBNST or alBNST, and greater significantly in orchiectomized males compared with gonadally intact males in ovBNST. The total number of neurons was also greater significantly in females than in males in ovBNST and alBNST, but it was not affected by GDX. Venus-expressing CRF neurons showed female-biased sexual dimorphism in ovBNST and alBNST of the mouse. Expression of Venus in these subnuclei was controlled by gonadal steroids."*

(Ponzi et al., 2020)

Effects of Prenatal Exposure to a Low-Dose of Bisphenol A on Sex Differences in Emotional Behavior and Central Alpha2-Adrenergic Receptor Binding

> *"Amid the first areas of the CNS that were found to be sexually dimorphic are regions regulating reproductive behaviors, such as the medial preoptic area (mPOA) and the anteroventral periventricular nucleus (AVPV) of the hypothalamus (Simerly, 2002). However, differences in the size of brain nuclei, number of cells and connectivity have been found in other regions of the brain not directly involved with reproductive behaviors, such as the hippocampus (Zhang et al., 2008), the locus coeruleus (LC) (Guillamón et al., 1988), the amygdala and other limbic regions (Simerly, 2002)(Cooke et al., 1999). The LC is a noradrenergic nucleus of the brainstem that is involved in arousal, vigilance, attention, cognition and the stress response (Berridge et al., 2003)(Charmandari et al., 2005) (Sara, 2009)"*

> *"The medial preoptic area (mPOA) of the hypothalamus is a sexually dimorphic area that regulates sex behaviors in males [28,29] and maternal behavior [30]."*

(Tsukahara et al., 2020)

Sexually Dimorphic Formation of the Preoptic Area and the Bed Nucleus of the Stria Terminalis by Neuroestrogens

> *"The preoptic area (POA) and the bed nucleus of the stria terminalis (BNST) are sexually dimorphic brain regions that are involved in the regulation of sex-specific social behaviors and the reproductive neuroendocrine system"*

> *"neuroestrogens induce the sex differences in glial and immune cells, which play an important role in the sexually dimorphic formation of the dendritic synapse patterning in the POA, and induce the sex differences in the cell number of specific neuronal cell groups in the POA and BNST"*

Neuronal Markers and Molecular Characteristics

Sexually dimorphic hypothalamic neurons express distinct molecular markers including kisspeptin, tyrosine hydroxylase, estrogen receptors, androgen receptors, and neuropeptides like PACAP and VIP. These markers show sex-specific expression patterns that enable differential responses to hormonal signals and coordinate sex-specific reproductive functions. (11 sources)

- **Kisspeptin (Kiss1)** neurons represent a major sexually dimorphic population, with AVPV/PeN Kiss1 neurons being more numerous in females than males and participating in estradiol-induced positive feedback control of GnRH secretion (Stephens et al., 2017) (Lee et al., 2022)
- **GPR54** (kisspeptin receptor) is expressed throughout the brain including hypothalamus, habenula, caudate nucleus, nucleus accumbens, and amygdala, and binds kisspeptins to directly regulate the reproductive

neuroendocrine axis (Kauffman et al., 2007) (Muir et al., 2001) (Kotani et al., 2001)

- **Tyrosine hydroxylase (TH)** neurons in the AVPV show pronounced sexual dimorphism, with females having 2-4 times more TH-immunoreactive neurons than males (Park et al., 2010) (Pandey et al., 2011)
- Most AVPV/PeN Kiss1 cells co-express tyrosine hydroxylase, the rate-limiting enzyme in dopamine synthesis (Stephens et al., 2017)

- **Estrogen receptor 1 (Esr1)** is highly expressed in MPOA neurons involved in both parental behaviors and male-type mating behaviors (Hashikawa et al., 2021)
- **Androgen receptor (Ar)** and **progesterone receptor (Pgr)** are enriched in the MPOA, where differential effects of circulating steroids between sexes underlie sexually dimorphic features (Hashikawa et al., 2021)
- Kisspeptin neurons express estrogen receptors starting in utero, allowing early hormonal regulation (Wartenberg et al., 2021)

- **PACAP (pituitary adenylate cyclase-activating polypeptide)** shows male-biased expression in POA neurons that project to the anterior pituitary (Yamashita et al., 2021)
- **VIP (vasoactive intestinal peptide)** exhibits female-biased expression in the same POA neuronal population, with both peptides regulated by adult estrogen secretion (Yamashita et al., 2021)
- **Galanin (Gal)** is expressed in MPOA neurons essential for parental behaviors (Hashikawa et al., 2021)
- **Neurotensin (Nts)** is found in MPOA neurons that govern female socio-sexual behaviors (Hashikawa et al., 2021)

- **Oxytocin receptor (OTR)** shows sexually dimorphic expression, with higher levels in males in the ventral premammillary nucleus (PMv) and higher expression in females in the AVPV at sexual maturity (Newmaster et al., 2019)

- **Aromatase** neurons form a network of approximately 6,000 neurons in the hypothalamus and amygdala, with sexually dimorphic clusters in the arcuate nucleus that convert testosterone to estrogen to regulate kisspeptin neuron activity (Wartenberg et al., 2021)

Paper	Targeted Neuronal Population	Key Molecular Markers Investigated	Brain Regions Of Focus	Function/ Phenotype Studied	Mechanism Of Regulation	Sexual Dimorphism Assessment
Effects of Selective Deletion of Tyrosine Hydroxylase from Kisspeptin Cells on Puberty and Reproduction in Male and Female Mice	AVPV/PeN and ARC Kiss1 neurons	Kiss1, tyrosine hydroxylase (TH)	hypothalamic AVPV/PeN and ARC nuclei	puberty onset, reproduction, body weights, hormone levels	dopamine signaling from AVPV/ PeN Kiss1 cells	explicitly investigates sexual dimorphism

Paper	Targeted Neuronal Population	Key Molecular Markers Investigated	Brain Regions Of Focus	Function/ Phenotype Studied	Mechanism Of Regulation	Sexual Dimorphism Assessment
Sexual Dimorphism in Kisspeptin Signaling	Hypothalamic Kisspeptin neurons (ARC, AVPV/PeN) and Gonadotropin-releasing hormone (GnRH) neurons.	Kisspeptin (KP) and kisspeptin receptor (KPR)	Hypothalamus (arcuate, anteroventral periventricular, periventricular preoptic nuclei), amygdala, hippocampus, cortex, brainstem, pituitary, pons, thalamus, striatum.	onset of puberty, gonadal function, GnRH secretion, GPN surge	GPR54 signaling, gonadal steroid feedback, transcriptional regulation, and neuropeptide co-expression.	Explicitly investigates sex differences in neuronal markers, molecular characteristics, and functional outcomes.
The Kisspeptin Receptor GPR54 Is Required for Sexual Differentiation of the Brain and Behavior	AVPV, Kiss1 mRNA-containing neurons, spino-bulbocavernosus nucleus motoneurons	GPR54, Kiss1 mRNA, tyrosine hydroxylase	anteroventral periventricular nucleus, spino-bulbocavernosus nucleus	sexual differentiation of the brain and behavior	Kisspeptin-GPR54 signaling	Yes, investigates sex differences in neuronal markers.
Male-like sexual behavior of female mouse lacking fucose mutarotase	anteroventral periventricular nucleus (AVPv) of the preoptic region	FucM, tyrosine hydoxylase, alpha-fetoprotein (AFP)	anteroventral periventricular nucleus (AVPv) of the preoptic region	Abnormal sexual receptivity, male-like sexual behavior	reduced fucosylation leading to reduced AFP level	Yes, reduction in tyrosine hydoxylase positive neurons in AVPv
Pubertal sex hormones control transcriptional trajectories in the medial preoptic area	Slc32a1 and Esr1 (Vgat+ Esr1+) neurons in MPOA	Slc32a1 and Esr1	medial preoptic area (MPOA)	Pubertal maturation, reproductive behaviors, mating behavior	Transcriptional dynamics controlled by testosterone and estradiol levels	Yes, sexually dimorphic gene-regulatory network and transcriptional progressions

Paper	Targeted Neuronal Population	Key Molecular Markers Investigated	Brain Regions Of Focus	Function/ Phenotype Studied	Mechanism Of Regulation	Sexual Dimorphism Assessment
Sexually Dimorphic Neurosteroid Synthesis Regulates Neuronal Activity in the Murine Brain	Kisspeptin neurons and aromatase neurons	Aromatase, kisspeptin, estrogen receptors	hypothalamus, amygdala, arcuate nucleus	Reproductive neural circuits and kisspeptin neuron activity	Neurosteroid synthesis, testosterone to estrogen conversion	Yes, sexually dimorphic aromatase neuronal network and activity.

Evidence

(Stephens et al., 2017)

Effects of Selective Deletion of Tyrosine Hydroxylase from Kisspeptin Cells on Puberty and Reproduction in Male and Female Mice

"Kiss1-syntheizing neurons reside primarily in the hypothalamic anteroventral periventricular (AVPV/PeN) and arcuate (ARC) nuclei. AVPV/PeN Kiss1 neurons are sexually dimorphic, with females expressing more Kiss1 than males, and participate in estradiol (E2)-induced positive feedback control of GnRH secretion. In mice, most AVPV/PeN Kiss1 cells coexpress tyrosine hydroxylase (TH), the rate-limiting enzyme in catecholamine synthesis (in this case, dopamine)."

(Lee et al., 2022)

Sexual Dimorphism in Kisspeptin Signaling

"Hypothalamic KPs and KPR display a high degree of sexual dimorphism in expression and function. KPs act on KPR in gonadotropin releasing hormone (GnRH) neurons and induce distinct patterns of GnRH secretion in males and females"

"Gonadal hormones inhibit the KP neurons within the arcuate nucleus and generate pulsatile GnRH mediated gonadotropin (GPN) secretion in both sexes. However, the numbers of KP neurons in the anteroventral periventricular nucleus and preoptic area are greater in females, which release a large amount of KPs in response to a high estrogen level and induce the preovulatory GPN surge."

(Kauffman et al., 2007)

The Kisspeptin Receptor GPR54 Is Required for Sexual Differentiation of the Brain and Behavior

"adult testosterone-treated GPR54 KO males displayed "female-like" numbers of tyrosine hydroxylase-immunoreactive and Kiss1 mRNA-containing neurons in the anteroventral periventricular nucleus and likewise possessed fewer motoneurons in the spino-bulbocavernosus nucleus than did WT males"

"GPR54 mRNA is expressed in the periphery (pancreas, placenta, and testes) and throughout the brain (hypothalamus, habenula, caudate nucleus, nucleus accumbens, and amygdala) (Lee et al., 1999)(Kotani et al., 2001)(Muir et al., 2001). GPR54 binds natural ligands called kisspeptins, which are encoded by the Kiss1 gene and synthesized in several brain regions, including the hypothalamic arcuate (ARC) and anteroventral periventricular (AVPV) nuclei (Kotani et al., 2001)(Muir et al., 2001)(Ohtaki et al., 2001)(Gottsch et al., 2004)(Shahab et al., 2005). GPR54 and kisspeptins are involved in the direct regulation of the reproductive neuroendocrine axis"

(Muir et al., 2001)

AXOR12, a Novel Human G Protein-coupled Receptor, Activated by the Peptide KiSS-1*

"A novel human G protein-coupled receptor named AXOR12, exhibiting 81% homology to the rat orphan receptor GPR54, was cloned from a human brain cDNA library. Heterologous expression of AXOR12 in mammalian cells permitted the identification of three surrogate agonist peptides, all with a common C-terminal amidated motif. High potency agonism, indicative of a cognate ligand, was evident from peptides derived from the gene KiSS-1, the expression of which prevents metastasis in melanoma cells. Quantitative reverse transcriptase-polymerase chain reaction was used to study the expression of AXOR12 and KiSS-1 in a variety of tissues. The highest levels of expression of AXOR12 mRNA were observed in brain, pituitary gland, and placenta. The highest levels ofKiSS-1 gene expression were observed in placenta and brain. A polyclonal antibody raised to the C terminus of AXOR12 was generated and used to show localization of the receptor to neurons in the cerebellum, cerebral cortex, and brainstem. The biological

significance of these expression patterns and the nature of the putative cognate ligand for AXOR12 are discussed."

(Kotani et al., 2001)

The Metastasis Suppressor Gene KiSS-1 Encodes Kisspeptins, the Natural Ligands of the Orphan G Protein-coupled Receptor GPR54*

"Natural peptides displaying agonist activity on the orphan G protein-coupled receptor GPR54 were isolated from human placenta. These 54-, 14,- and 13-amino acid peptides, with a common RF-amide C terminus, derive from the product of KiSS-1, a metastasis suppressor gene for melanoma cells, and were therefore designated kisspeptins. They bound with low nanomolar affinities to rat and human GPR54 expressed in Chinese hamster ovary K1 cells and stimulated PIP2 hydrolysis, Ca2+ mobilization, arachidonic acid release, ERK1/2 and p38 MAP kinase phosphorylation, and stress fiber formation but inhibited cell proliferation. Human GPR54 was highly expressed in placenta, pituitary, pancreas, and spinal cord, suggesting a role in the regulation of endocrine function. Stimulation of oxytocin secretion after kisspeptin administration to rats confirmed this hypothesis."

(Park et al., 2010)

Male-like sexual behavior of female mouse lacking fucose mutarotase

"The populations of tyrosine hydroxylase (TH)-expressing hypothalamic cells are sexually dimorphic in rats and mice (Simerly, 1989) (Simerly et al., 1985). TH-immunoreactive (ir) positive neurons in the anteroventral periventricular nucleus (AVPv) of the preoptic region regulating gonadotropin release are necessary for female ovulation, whose numbers in a female, in addition to their size difference, are two to four times greater than in a male (Simerly et al., 1985)(Simerly, 1989)(Simerly et al., 1985)(Wiegand et al., 1982). Ovariectomy of the female does not affect TH expression, while chronic treatment with high levels of estradiol reduces its expression (Simerly et al., 1985)(Simerly, 1989)."

(Pandey et al., 2011)

The Israel Society for Neuroscience 19th Annual Meeting Eilat, Dec. 12 - 14, 2010 www.isfn.org.il

"The anteroventral periventricular nucleus of the preoptic region of the hypothalamus (AVPV) contains a sexually dimorphic population of neurons that are known to be involved in regulation of reproductive functions such as preovulatory gonadotropin surge in female rodents and possible sex differences in puberty onset. One of the sexual dimorphic populations is tyrosine hydroxylase (TH) positive neurons which their number is grater in females compare to males."

(Hashikawa et al., 2021)

Pubertal sex hormones control transcriptional trajectories in the medial preoptic area

"Molecularly defined subpopulations of neurons expressing a variety of neuropeptides and/or hormonal receptors in the MPOA are tightly associated with reproductive behaviors. MPOA neurons expressing Gal (galanin) or Esr1 (estrogen receptor 1) are essential for parental behaviors, while MPOA neurons expressing Esr1 or Nts (neurotensin) govern male-type mating behaviors and female socio-sexual behaviors, respectively"

"Since the MPOA is enriched in the expression of steroid hormone receptors genes (e.g., Esr1, androgen receptor (Ar), progesterone receptor (Pgr))"

"Differential effects of circulating steroids between sexes e.g., estrogen in females and testosterone in males, potentially underly sexually dimorphic features of the MPOA."

(Wartenberg et al., 2021)

Sexually Dimorphic Neurosteroid Synthesis Regulates Neuronal Activity in the Murine Brain

"Sex steroid hormones act on hypothalamic kisspeptin neurons to regulate reproductive neural circuits in the brain. Kisspeptin neurons start to express estrogen receptors in utero"

"We identified an aromatase neuronal network comprising ≈ 6000 neurons in the hypothalamus and amygdala. By birth, this network has become sexually dimorphic in a cluster of aromatase neurons in the arcuate nucleus adjacent to kisspeptin neurons. We demonstrate that male arcuate aromatase neurons convert testosterone to estrogen to regulate kisspeptin neuron activity."

(Yamashita et al., 2021)

Estrogen mediates sex differences in preoptic neuropeptide and pituitary hormone production in medaka

"two neuropeptides belonging to the secretin family, pituitary adenylate cyclase-activating polypeptide (Pacap) and vasoactive intestinal peptide (Vip), exhibit opposite patterns of sexually dimorphic expression in the same population of POA neurons that project to the anterior pituitary: Pacap is male-biased, whereas Vip is female-biased"

"the sexually dimorphic Pacap-and Vipexpressing neurons in the POA directly project to the anterior pituitary, where functional receptors for these peptides are expressed"

"Pacap and Vip produced therein act as hypophysiotropic factors"

"the sexually dimorphic expression of both Pacap and Vip is caused by adult secretion of estrogen"

(Newmaster et al., 2019)

Quantitative cellular-resolution map of the oxytocin receptor in postnatally developing mouse brains

"Sexual dimorphism of OTR expression. OTR is expressed in a sexually dimorphic manner as a part of neural circuit mechanism to generate behavioral differences in males and females 34,35 . Therefore, we compared OTR-Venus expression in male and female mice (N = 5 in male and female brains at different ages) to determine if there were any regions showing strong sexual dimorphism. Across the entire brain region throughout the postnatal development, we found significant sexual dimorphism in two hypothalamic regions (Fig. 7). The ventral premammillary nucleus (PMv) showed significantly higher OTR expression in males compared to females between P14 and P56 (Fig. 7a-d). In contrast, the anteroventral periventricular nucleus (AVPV) near the medial preoptic area showed higher OTR expression in females than males at P56, but not before (Fig. 7e-h). A recent study identified abundant estrogen-dependent OTR-expressing cells in the AVPV, co-expressing estrogen receptor in female mice 36 . This result suggests a potential role of OTR in sexual behavior [36][37][38] ."

Functional Roles in Reproductive Neuroendocrine Control

Sexually dimorphic hypothalamic neurons coordinate reproductive functions through distinct mechanisms: AVPV kisspeptin neurons mediate estrogen-positive feedback for ovulation in females, while arcuate kisspeptin neurons generate pulsatile GnRH release in both sexes. These populations show sex-specific firing patterns and connectivity that enable differential control of reproductive behaviors and hormone secretion. (10 sources)

The functional organization of sexually dimorphic reproductive circuits involves distinct neuronal populations that coordinate different aspects of reproductive neuroendocrine control. Kisspeptin neurons act as strong regulators of the hypothalamic-pituitary-gonadal axis, with their sexually dimorphic distribution enabling sex-specific patterns of gonadotropin-releasing hormone (GnRH) secretion (Zdon et al., 2024) (Zdon et al., 2025). The timing and activation of these hypothalamic neuronal populations is critical for pubertal onset, though the precise mechanisms controlling their activation remain poorly understood (Pereira et al., 2018).

Regional Functional Specialization

The two major kisspeptin populations exhibit distinct functional roles that reflect their sexually dimorphic organization. AVPV/PeN kisspeptin neurons in females mediate estrogen positive feedback to induce the preovulatory GnRH/LH surge, with these neurons showing greater numbers in females and releasing large amounts of kisspeptins in response to high estrogen levels (Nejad et al., 2017) (Lee et al., 2022) (Adachi et al., 2007). In contrast, arcuate nucleus kisspeptin neurons generate pulsatile GnRH-mediated gonadotropin secretion in both sexes through inhibition by gonadal hormones, with this population showing no sexual dimorphism (Nejad et al., 2017) (Lee et al., 2022).

Electrophysiological Properties and Sex Differences

The functional differences between sexes are reflected in distinct electrophysiological properties of kisspeptin neurons. AVPV/PeN kisspeptin neurons display sexually dimorphic biophysical characteristics, with male neurons exhibiting higher action potential frequency and receiving more excitatory synaptic transmission compared to females (Frazao et al., 2013). These differences in cellular activity may underlie the striking sex differences in GnRH secretion patterns, while arcuate kisspeptin neurons show heterogeneous activity patterns without sexual dimorphism (Frazao et al., 2013).

Steroid Hormone Integration

Sexually dimorphic neuronal populations in the rostral hypothalamus serve as critical integration sites for steroid hormone signals. Estradiol-sensitive neurons in the periventricular nucleus and medial preoptic nucleus show pronounced sexual dimorphism, with females exhibiting significantly higher percentages of dual-labeled cells that may directly relay steroid hormone signals to regulate GnRH secretion (Moore et al., 2019). Beyond the hypothalamus, sexually dimorphic kisspeptin populations in limbic regions like the bed nucleus of the stria terminalis and lateral septum integrate reproductive axis function with social behaviors

and metabolic functions (Zdon et al., 2024) (Zdon et al., 2025).

Behavioral Circuit Organization

The sexually dimorphic neural circuits underlying reproductive behaviors involve complex pathways that integrate sensory input with hormonal status. Sexual behavior activation relies on pheromone detection through chemosensory pathways that project through the olfactory bulbs to the medial amygdala, which then connects to sexually dimorphic hypothalamic regions (Torres et al., 2024). In males, the medial preoptic area plays a critical role in sexual behavior expression with projections to spinal centers controlling erection and ejaculation, while in females, the ventromedial hypothalamus is essential for lordosis behavior through connections to brainstem and spinal motor circuits (Torres et al., 2024) (Johnson et al., 2021).

Evidence

(Zdon et al., 2024)

7996 Neuroanatomic Characterization of Mouse Kisspeptin Neurons in the Bed Nucleus of the Stria Terminalis

> *"Kisspeptin (Kiss1) neurons are essential for reproductive function as they are strong regulators of the hypothalamic-pituitary-gonadal axis (HPG). Mammalian Kiss1 neurons are mainly located in the hypothalamus, controlling gonadotropin-releasing hormone (GnRH) secretion, where the anteroventral periventricular nucleus (AVPV) population is morphologically and functionally sexually differentiated. Kiss1 neurons are also found in limbic areas, such as the bed nucleus of the stria terminalis (BNST). This nucleus is involved in the integration of the HPG axis with sex-dependent social behaviors and metabolic functions"*
>
> *"male mice have significantly more Kiss1prBNST neurons than females (Male: 139.4 ± 23.32 vs. Female: 43.70 ± 5.33 neurons; **P < 0.01). This sex-biased distribution was absent in Kiss1aBNST neurons"*
>
> *"fibers from aBNST-injected Kiss1 neurons projecting widely to areas that include the AVPV, medial preoptic area (MPO), lateral preoptic area (LPO), and periventricular hypothalamic nucleus (PVH). Moreover, fibers and cell bodies from male prBNST-injected Kiss1 neurons projected to dorsal lateral BNST (dlBNST), juxta-periventricular lateral hypothalamic area (JPLH), and the posterolateral stria terminalis (STMPL). This study observed a sexual dimorphic Kiss1 neuronal population in the prBNST"*

(Zdon et al., 2025)

SUN-088 Sexual Dimorphism and Social Behavioral Role of Lateral Septal Kisspeptin Neurons in Male Mice

> *"Kisspeptin (Kiss1) neurons are crucial regulators of the hypothalamic-pituitary-gonadal axis (HPG), making them essential for reproductive function. While primarily located in the hypothalamus where they control gonadotropin-releasing hormone (GnRH) secretion, additional Kiss1 neurons are found in areas outside of the hypothalamus, particularly in limbic areas, such as the lateral septum (LS). The LS, comprising ventral (VLS) and intermediate (ILS) divisions, integrates social, feeding, anxiety, and fear behaviors, and Kiss1 neurons in this region are regulated by sex steroids"*
>
> *"Quantitative analysis revealed a marked sexual dimorphism, with females showing significantly more Kiss1ILS compared to males (Male: 7.00 ± 2.864 vs. Female: 74.40 ± 17.68 neurons, **P = 0.0043). This sexual dimorphism was absent in Kiss1VLS neurons."*

(Pereira et al., 2018)

Timing mechanism of sexually dimorphic nervous system differentiation

> *"The onset of puberty in mammals is characterized by the activation of neurons in the hypothalamus that produce hormonal signals (Avendano et al., 2017). However, the mechanisms that control the activation of these neurons and hence the timing of pubertal onset are poorly understood."*

(Nejad et al., 2017)

The Role of Kisspeptin in Female Reproduction

> *"In mammals the anatomy and physiology of the neuroendocrine reproductive axis differs between the genders"*
>
> *"The Kisspeptin system is apparently critical for brain gender differentiation, acting through the regulation of postnatal T secretion. Distribution of Kisspeptin neurons in the hypothalamus varies between species. In mammals there are 2 major regions of these neurons; a rostral one in the Pre-Optic Area (POA) and a caudal one in the arcuate nucleus, with proportionally more Kisspeptin neurons in the ARC than in the POA region (4,30). In rodents, the POA regions are concentrated in the Anteroventral Periventricular Nucleus (AVPV). Anatomical differences between genders have been reported in the hypothalamus of some species, e.g. the rat AVPV is sexually dimorphic, with a greater number of KISS1 neurons in females compared to males (30). Like most sex differences in the brain, this sexual dimorphism is likely caused during the perinatal critical period by exposure to testosterone (or its*

metabolites) (31)"

"KISS1 neurons express sex steroid receptors and are regulated by gonadal sex steroids, mediating the effects of estrogen on GnRH neurons"

"In female rodents, the AVPV mediates E2 positive feedback, causing occurrence of pre-ovulatory GnRH/LH surge. Circadian cues also impinge upon AVPA, whereby KISS1 neurons in the AVPV receiving hormonal and temporal signals give rise to timely LH surge (30). In contrast to AVPV, the arcuate nucleus shows no differences between males and females"

(Lee et al., 2022)

Sexual Dimorphism in Kisspeptin Signaling

"Hypothalamic KPs and KPR display a high degree of sexual dimorphism in expression and function. KPs act on KPR in gonadotropin releasing hormone (GnRH) neurons and induce distinct patterns of GnRH secretion in males and females"

"Gonadal hormones inhibit the KP neurons within the arcuate nucleus and generate pulsatile GnRH mediated gonadotropin (GPN) secretion in both sexes. However, the numbers of KP neurons in the anteroventral periventricular nucleus and preoptic area are greater in females, which release a large amount of KPs in response to a high estrogen level and induce the preovulatory GPN surge."

(Adachi et al., 2007)

Involvement of anteroventral periventricular metastin/kisspeptin neurons in estrogen positive feedback action on luteinizing hormone release in female rats.

"Metastin/kisspeptin, the KiSS-1 gene product, has been identified as an endogenous ligand of GPR54 that reportedly regulates GnRH/LH surges and estrous cyclicity in female rats. The aim of the present study was to determine if metastin/kisspeptin neurons are a target of estrogen positive feedback to induce GnRH/LH surges. We demonstrated that preoptic area (POA) infusion of the anti-rat metastin/kisspeptin monoclonal antibody blocked the estrogen-induced LH surge, indicating that endogenous metastin/kisspeptin released around the POA mediates the estrogen positive feedback effect on GnRH/LH release. Metastin/kisspeptin neurons in the anteroventral periventricular nucleus (AVPV) may be responsible for mediating the feedback effect because the percentage of c-Fos-expressing KiSS-1 mRNA-positive cells to total KiSS-1 mRNA-positive cells was significantly higher in the afternoon than in the morning in the anteroventral periventricular nucleus (AVPV) of high estradiol (E(2))-treated females. The percentage of c-Fos-expressing metastin/kisspeptin neurons was not different between the afternoon and morning in the arcuate nucleus (ARC). Most of the KiSS-1 mRNA expressing cells contain ERalpha immunoreactivity in the AVPV and ARC. In addition, AVPV KiSS-1 mRNA expressions were highest in the proestrous afternoon and lowest in the diestrus 1 in females and were increased by estrogen treatment in ovariectomized animals. On the other hand, the ARC KiSS-1 mRNA expressions were highest at diestrus 2 and lowest at proestrous afternoon and were increased by ovariectomy and decreased by high estrogen treatment. Males lacking the surge mode of GnRH/LH release showed no obvious cluster of metastin/kisspeptin-immunoreactive neurons in the AVPV when compared with high E(2)-treated females, which showed a much greater density of these neurons. Taken together, the present study demonstrates that the AVPV metastin/kisspeptin neurons are a target of estrogen positive feedback to induce GnRH/LH surges in female rats."

(Frazao et al., 2013)

Shift in Kiss1 Cell Activity Requires Estrogen Receptor α

"Kiss1 neurons in the anteroventral periventricular and anterior periventricular nuclei (AVPV/PeN) of males and females exhibit a bimodal resting membrane potential (RMP) influenced by KATP channels, suggesting the presence of two neuronal populations defined as type I (irregular firing patterns) and type II (quiescent). Kiss1 neurons in the arcuate nucleus (Arc) are also composed of firing and quiescent cells, but unlike AVPV/PeN neurons, the range of RMPs did not follow a bimodal distribution. Moreover, Kiss1 neuronal activity in the AVPV/PeN, but not in the Arc, is sexually dimorphic"

"The AVPV is a sexually dimorphic site with a differential distribution pattern of several neurotransmitters and neuropeptides, including kisspeptin"

"the AVPV/PeN Kiss1 neurons also displayed sexually dimorphic biophysical properties, such that neurons of males exhibited higher activity (i.e., higher APs frequency) and received higher excitatory synaptic transmission (i.e., higher sEPSC frequency)"

"differences in Kiss1 cellular activity may underlie the striking difference in GnRH secretion between sexes"

"Male and female Arc Kiss1 neurons showed a heterogeneous pattern of cellular activity, but no bimodal distribution of RMPs. No evidence for a sexually dimorphic pattern of Arc Kiss1 cellular activity was noticed, which is in line with a lack of difference in the distribution of Kiss1 mRNA in the Arc between sexes"

(Moore et al., 2019)

SAT-426 Rabies-Mediated Monosynaptic Tract-Tracing of Sexually Dimorphic Estrogen-Sensitive Afferents to KNDy Neurons in the Mouse

"Kisspeptin/Neurokinin B/Dynorphin (KNDy) neurons of the arcuate nucleus (ARC) are critical for reproductive status and implicated to relay a number of peripheral signals, including gonadal steroids, to regulate GnRH secretion"

"Dual-labeled cells were detected within the bed nucleus of the stria terminalis and the hypothalamic rostral periventricular area, medial preoptic nucleus (MPON), ventromedial and anterior hypothalamus, ARC and premammillary nucleus. Compared to male mice, females exhibited a significantly higher percentage of dual-labeled cells in the periventricular nucleus (PeN, 46.7±16.4% in females, 7.8±5.1% in males, p<0.05) and MPON (34.3±7.2% in females, 10.9±6% in males, p<0.05). The PeN and MPON are critical for reproductive endocrine function and sexual behavior in the mouse, respectively. Together, these data indicate that sexually dimorphic estradiol-sensitive neuronal populations in the rostral hypothalamus may directly relay steroid hormone signals to KNDy neurons to regulate GnRH secretion and perhaps other functions."

(Torres et al., 2024)

Reproductive function and behaviors: an update on the role of neural estrogen receptors alpha and beta

"Sexual behavior relies on the activation of complex, sexually dimorphic neural circuits (10) (Figure 2). It is triggered by the detection of sensory stimuli, especially pheromones, that are detected by chemosensory neurons located in the vomeronasal organ and the main olfactory epithelium (11,12). These neurons project to the main olfactory bulb (MOB) and the accessory olfactory bulb (AOB), which innervates among other structures, the medial amygdala (MeA) (13,14). Neurons from the MeA then project to the bed nucleus of the stria terminalis (BNST), the medial preoptic area (mPOA), and the ventromedial nucleus of the hypothalamus (VMH). In males, the mPOA plays a critical role in the expression of sexual behavior. Projections from the hypothalamic paraventricular nucleus (PVN) are sent to the spinal centers that promote erection and ejaculation (15,16). In females, the VMH is essential for the activation of sexual behavior. It sends projections to the periaqueductal gray matter (PAG) that project to the brainstem and spinal cord, which innervate the axial muscles involved in the lordosis posture (17)."

(Johnson et al., 2021)

Posterodorsal Medial Amygdala Regulation of Female Social Behavior: GABA versus Glutamate Projections

"Lordosis, the measure of female sexual receptivity, is one of the most apparent sexually dimorphic reproductive behaviors. Lordosis is regulated by estrogen and progesterone (P4) acting within a hypothalamic-limbic circuit, consisting of the arcuate, medial preoptic, and ventromedial nuclei of the hypothalamus. Social cues are integrated into the circuit through the amygdala. The posterodorsal part of the medial amygdala (MeApd) is involved in sexually dimorphic social and reproductive behaviors, and sends projections to hypothalamic neuroendocrine regions. GABA from the MeApd appears to facilitate social behaviors, while glutamate may play the opposite role."

Connectivity Patterns and Synaptic Organization

Sexually dimorphic hypothalamic neurons show distinct connectivity patterns despite similar anatomical distributions, with females receiving more inputs from estrogen-sensitive regions and exhibiting earlier synaptic development. These sex-specific wiring patterns enable differential integration of reproductive and behavioral signals through specialized synaptic networks. (6 sources)

The synaptic organization of sexually dimorphic reproductive circuits reveals fundamental sex differences in neural connectivity that extend beyond simple anatomical dimorphism. Even when neuronal populations show similar distributions between sexes, their underlying connectivity patterns can be markedly different, creating the substrate for sexually dimorphic function (Esteves et al., 2019). This principle is exemplified by neuroendocrine dopamine (NEDA) neurons in the arcuate nucleus, which despite having monomorphic distribution patterns, receive sexually dimorphic inputs from multiple hypothalamic regions including the anterior hypothalamic nucleus, anteroventral periventricular nucleus, medial preoptic nucleus, paraventricular hypothalamic nucleus, and suprachiasmatic nucleus (Esteves et al., 2019).

Sex-Specific Input Patterns to Reproductive Neurons

Detailed monosynaptic tracing studies reveal that female-dominant sex differences in afferent connectivity are particularly pronounced for inputs from estrogen-sensitive hypothalamic nuclei. KNDy neurons (kisspeptin/neurokinin B/dynorphin-expressing cells) in the arcuate nucleus receive over 90% of their monosynaptic input from hypothalamic nuclei in both sexes, but females show significantly higher percentages of afferents from the rostral periventricular area (RP3V), specifically within the periventricular nucleus (Moore et al._1, 2019). Importantly, a greater percentage of these afferent cells co-express estrogen receptor alpha in females compared to males, suggesting enhanced capacity for estrogen-mediated synaptic

regulation in female circuits (Moore et al._1, 2019).

Developmental Sexual Dimorphism in Synaptic Formation

The establishment of sexually dimorphic connectivity patterns begins during early development through sex-specific differences in neuritic differentiation and synaptic formation. Female hypothalamic neurons differentiate earlier than male neurons in culture and show higher expression of the neuritogenic factor Ngn3, while male neurons exhibit retarded neuritic differentiation (Scerbo et al., 2014). This developmental timing difference means female neurons establish a higher number of postsynaptic sites when early synaptic afferents arrive, potentially creating fundamentally different patterns of synaptic connectivity between the sexes (Scerbo et al., 2014).

Integration with Dopaminergic and Neuropeptide Systems

Sexually dimorphic brain regions provide extensive synaptic input to neuromodulatory systems that coordinate reproductive behaviors. Dopamine-producing neurons in the ventral tegmental area and other brain regions receive direct synaptic input from sexually dimorphic structures including the medial preoptic area, bed nucleus of the stria terminalis, and medial amygdala (Chung et al., 2017) (Chung et al._1, 2017). Similarly, both oxytocinergic and vasopressinergic neuronal systems show sex differences in their projection patterns, with vasopressin neurons displaying more pronounced sexually dimorphic outputs despite sharing highly similar and sex-conserved synaptic inputs with oxytocin neurons (Freda et al., 2022).

Evidence

(Esteves et al., 2019)

Sexually dimorphic neuronal inputs to the neuroendocrine dopaminergic system governing prolactin release

> *"we mapped whole-brain monosynaptic NEDA inputs in male and female mice. Accordingly, we employed a rabies virus based monosynaptic tracing system capable of retrogradely mapping inputs into genetically defined neuronal populations. To gain genetic access to NEDA neurones, we used the dopamine transporter promoter. Here, we unravel 59 brain regions that synapse onto NEDA neurones and reveal that male and female mice, despite monomorphic distribution of NEDA neurones in the arcuate nucleus of the hypothalamus, receive sexually dimorphic amount of inputs from the anterior hypothalamic nucleus, anteroventral periventricular nucleus, medial preoptic nucleus, paraventricular hypothalamic nucleus, posterior periventricular nucleus, supraoptic nucleus, suprachiasmatic nucleus, lateral supramammillary nucleus, tuberal nucleus and periaqueductal grey. Beyond highlighting the importance of considering sex as a biological variable when evaluating connectivity in the brain, these results illustrate a case where a neuronal population with similar anatomical distribution has a subjacent sexually dimorphic connectivity pattern, potentially capable of contributing to the sexually dimorphic nature of PRL release and function."*

(Moore et al._1, 2019)

Kisspeptin/Neurokinin B/Dynorphin (KNDy) cells as integrators of diverse internal and external cues: evidence from viral-based monosynaptic tract-tracing in mice

> *"Neurons in the hypothalamic arcuate nucleus (ARC) that co-express kisspeptin, neurokinin B and dynorphin (KNDy cells) are essential for mammalian reproduction as key regulators of gonadotropin-releasing hormone (GnRH) secretion"*
>
> *"Over 90% of monosynaptic input to KNDy neurons originated from hypothalamic nuclei in both male and female mice. The greatest input arose from non-KNDy ARC neurons, including proopiomelanocortin-expressing cells. Significant female-dominant sex differences in afferent input were detected from estrogen-sensitive hypothalamic nuclei critical for reproductive endocrine function and sexual behavior in mice, indicating KNDy cells may provide a unique site for the coordination of sex-specific behavior and gonadotropin release"*
>
> *"A higher percentage of afferents to KNDy cells were detected in the RP3V of females compared to males, specifically within the PeN. Further, of the afferent cells detected in the PeN, a significantly greater percentage co-expressed ERα in female mice compared to males. The RP3V contains multiple estrogen-sensitive populations 59, including a greater estrogen-sensitive rostral kisspeptin population (RP3V kisspeptin neurons) in females compared to males that is implicated in mediating the preovulatory surge in GnRH/LH release in response to high estradiol levels in females."*

(Scerbo et al., 2014)

Neurogenin 3 mediates sex chromosome effects on the generation of sex differences in hypothalamic neuronal development

"Our findings have revealed a sex dimorphic expression of Ngn3 in hypothalamic neurons. Male neurons, which show a retarded neuritic differentiation compared to female neurons, showed decreased mRNA levels of the neuritogenic factor Ngn3"

"female hypothalamic primary neurons differentiated earlier than male neurons in culture"

"female neurons would show a higher number of postsynaptic sites available when early synaptic afferents arrive and could then, by this mechanism, establish a different pattern of synaptic connectivity than male neurons"

"some parameters of the neuritogenic process are sexually dimorphic and others are sex-independent."

(Chung et al., 2017)

Sexual congruency in the connectome and translatome of VTA dopamine neurons

"dopamine producing neurons located in numerous brain regions including the ventral midbrain, hypothalamus, olfactory bulb, and retina"

"dopamine neurons receive direct synaptic input from brain regions previously demonstrated to be sexually dimorphic in their organization, including the medial preoptic area (MPA), bed nucleus of the stria terminalis (BNST), and the medial amygdala"

(Chung et al._1, 2017)

Sexual congruency in the connectome and translatome of VTA dopamine neurons

"dopamine producing neurons of the ventral midbrain located in the VTA"

"dopamine neurons receive direct synaptic input from brain regions previously demonstrated to be sexually dimorphic in their organization, including the medial preoptic area (MPA), bed nucleus of the stria terminalis (BNST), and the medial amygdala (Simerly, 2002)"

(Freda et al., 2022)

Brainwide input-output architecture of paraventricular oxytocin and vasopressin neurons

"neurohypophyseal peptides may represent a single integrative neuromodulatory system, yet both peptides subserve sexually dimorphic functions at the behavioral level"

"Sex differences in the central circuitry of the oxytocin and vasopressin systems may underlie sex-variant behaviors, but it is currently unknown whether the synaptic inputs or outputs of each neuropeptidergic system vary across males and females"

"We observed that both oxytocinergic and vasopressinergic neurons share highly similar synaptic inputs that are sex-conserved. Projection patterns differed across systems and showed sex differences, more pronounced in the vasopressin neurons."

Developmental and Hormonal Regulation

Sexual dimorphism in hypothalamic neurons emerges through complex interactions between sex chromosomes, hormonal signals, and developmental timing, with female neurons showing earlier differentiation and enhanced neuritogenic factor expression. These developmental differences establish distinct neural architectures that enable sex-specific responses to reproductive hormones and metabolic signals throughout life. (8 sources)

The development of sexually dimorphic hypothalamic circuits involves intricate interactions between genetic, hormonal, and environmental factors that shape neural architecture during critical developmental periods. Sex chromosome complement plays a fundamental role independent of hormonal influences, with XX hypothalamic neurons showing enhanced neuritogenic capacity compared to XY neurons regardless of gonadal sex (Zapata et al., 2021). These chromosomal effects manifest through differential expression of X-linked genes including Kdm6a, Eif2s3x, and Ddx3x, which show consistently higher expression in XX compared to XY neurons throughout development from embryonic day 14 through adulthood (Zapata et al., 2021) (Xu et al., 2008).

Early Developmental Programming

The establishment of sexual dimorphism begins early in development through sex-specific differences in neuronal growth characteristics and gene expression patterns. Female (XX) hypothalamic neurons demonstrate faster growth rates, develop longer axons, and express higher levels of the neuritogenic factor neurogenin 3 (Ngn3) compared to male (XY) neurons before perinatal masculinization occurs (Zapata et al.,

2021) (Cisternas et al., 2020). These early differences in neuritic development establish the organizational framework for hypothalamic neuronal projections, ensuring precise structural and functional patterning that is crucial for proper circuit formation underlying sex-related behaviors (Cisternas et al., 2020).

Hormonal Influences on Circuit Development

Hormonal signals during critical developmental windows generate permanent sex differences in hypothalamic neuronal circuits controlling neuroendocrine events, feeding, growth, reproduction, and behavior (Marraudino et al., 2019). The ventromedial hypothalamus, which contains heterogeneous populations of estrogen receptor alpha-expressing neurons with different projections and sex-related functions, exemplifies how hormonal programming shapes circuit organization (Cisternas et al., 2020) (Kim et al., 2019) (Correa et al., 2015). Environmental factors can also influence this developmental process, as demonstrated by phytoestrogens like genistein, which exert sexually dimorphic actions on hypothalamic neuron development and alter normal patterns of sex differences in neuritogenesis (Marraudino et al., 2019).

Metabolic Circuit Sexual Dimorphism

The hypothalamic circuits controlling energy homeostasis exhibit pronounced sexual dimorphism that emerges through developmental regulation of key neuronal populations. Ngn3 serves as a developmental regulator of energy homeostasis systems in the hypothalamus, pancreas, and intestine, with POMC-positive and NPY-positive hypothalamic neurons forming central microcircuits for integrating peripheral energy status signals and controlling feeding and energy expenditure (Zapata et al., 2022). These metabolic control systems show notable sexual dimorphism, with the melanocortin (MSH) system demonstrating increased leptin responsiveness and decreased insulin responsiveness in females compared to males, while NPY/AgRP circuits exhibit sexually dimorphic expression patterns in the arcuate nucleus, dorsomedial hypothalamus, and paraventricular nucleus (Street et al., 2018).

Evidence

(Zapata et al., 2021)

X-linked histone H3K27 demethylase Kdm6a regulates sexually dimorphic differentiation of hypothalamic neurons

> *"X-linked genes Kdm6a, Eif2s3x and Ddx3x showed higher expression in XX compared to XY neurons, regardless of gonadal sex"*
>
> *"XX hypothalamic neurons grow faster, have longer axons, and exhibit higher expression of the neuritogenic gene neurogenin 3 (Ngn3) than XY before perinatal masculinization"*
>
> *"Kdm6a expression pattern with higher mRNA levels in XX than XY did not change with age at E14, P0, and P60 in hypothalamus"*
>
> *"Xu et al. (Xu et al., 2008) have reported higher levels of Kdm6a mRNA in adult females in cortex, striatum, suprachiasmatic nucleus, VMH, hippocampus (CA1/CA3), dentate gyrus, and habenula"*
>
> *"FCG XX hypothalamic neurons express higher Ngn3 mRNA levels than XY neurons, regardless of gonadal sex"*

(Xu et al., 2008)

Sex-Specific Differences in Expression of Histone Demethylases Utx and Uty in Mouse Brain and Neurons

> *"Although X inactivation is thought to balance gene expression between the sexes, some genes escape inactivation, potentially contributing to differences between males and females. Utx (ubiquitously transcribed tetratricopeptide repeat gene on X chromosome) is an escapee gene that encodes a demethylase specific for lysine 27 of histone H3, a mark of repressed chromatin. We found Utx to be expressed higher in females than in males in developing and adult brains and in adult liver. XX mice had a higher level of Utx than XY mice, regardless of whether they had testes or ovaries, indicating that the sexually dimorphic gene expression was a consequence of the sex chromosome complement. Females had significantly higher levels of Utx than males in most brain regions except in the amygdala. The regional expression of the Y-linked paralogue Uty (ubiquitously transcribed tetratricopeptide repeat gene on Y chromosome) was somewhat distinct from that of Utx, specifically in the paraventricular nucleus of the hypothalamus (high Uty) and the amygdala (high Utx), implying that the two paralogues may be differentially regulated. Higher expression of Utx compared with Uty was detected in P19 pluripotent embryonic carcinoma cells as well as in P19-derived neurons. This transcriptional divergence between the two paralogues was associated with high levels of histone H3 lysine 4 dimethylation at the Utx promoter and of histone H4 lysine 16 acetylation throughout the gene body, which suggests that epigenetic mechanisms*

control differential expression of paralogous genes."

(Cisternas et al., 2020)

Estradiol-dependent axogenesis and Ngn3 expression are determined by XY sex chromosome complement in hypothalamic neurons

"Hypothalamic neurons show sex differences in neuritogenesis, female neurons have longer axons and higher levels of the neuritogenic factor neurogenin 3 (Ngn3) than male neurons in vitro"

"the ventromedial hypothalamus which contains a highly heterogeneous population of ERα-expressing neurons with different projections and sex-related functions [43][44][45]. The sex differences in neuritic development may establish the organization of hypothalamic neuronal projections to ensure precise structural and functional patterning, which is crucial for proper circuit formation of sex-related behaviors"

"Ngn3 promotes the development of ventromedial SF1 neurons 16, and given the role of sex chromosome complement in regulating hormonal and non-hormonal expression of Ngn3, these results suggest that sex chromosomes might participate in the generation of neuronal populations controlling sexual development and reproduction in mice."

(Marraudino et al., 2019)

Sexually Dimorphic Effect of Genistein on Hypothalamic Neuronal Differentiation in Vitro

"This hormone generates sex differences in the development of hypothalamic neuronal circuits controlling neuroendocrine events, feeding, growth, reproduction and behavior"

"developmental actions of the phytoestrogen genistein result in permanent sexually dimorphic effects in some behaviors and neural circuits in vivo"

"genistein exerts sexually dimorphic actions on the development of hypothalamic neurons, altering the normal pattern of sex differences in neuritogenesis."

(Kim et al., 2019)

Multimodal Analysis of Cell Types in a Hypothalamic Node Controlling Social Behavior.

"The ventrolateral subdivision of the ventromedial hypothalamus (VMHvl) contains ≈ 4,000 neurons that project to multiple targets and control innate social behaviors including aggression and mounting. However, the number of cell types in VMHvl and their relationship to connectivity and behavioral function are unknown. We performed single-cell RNA sequencing using two independent platforms-SMART-seq (≈ 4,500 neurons) and 10x (≈ 78,000 neurons)-and investigated correspondence between transcriptomic identity and axonal projections or behavioral activation, respectively. Canonical correlation analysis (CCA) identified 17 transcriptomic types (T-types), including several sexually dimorphic clusters, the majority of which were validated by seqFISH. Immediate early gene analysis identified T-types exhibiting preferential responses to intruder males versus females but only rare examples of behavior-specific activation. Unexpectedly, many VMHvl T-types comprise a mixed population of neurons with different projection target preferences. Overall our analysis revealed that, surprisingly, few VMHvl T-types exhibit a clear correspondence with behavior-specific activation and connectivity."

(Correa et al., 2015)

An estrogen-responsive module in the ventromedial hypothalamus selectively drives sex-specific activity in females.

"Estrogen-receptor alpha (ERα) neurons in the ventrolateral region of the ventromedial hypothalamus (VMHVL) control an array of sex-specific responses to maximize reproductive success. In females, these VMHVL neurons are believed to coordinate metabolism and reproduction. However, it remains unknown whether specific neuronal populations control distinct components of this physiological repertoire. Here, we identify a subset of ERα VMHVL neurons that promotes hormone-dependent female locomotion. Activating Nkx2-1-expressing VMHVL neurons via pharmacogenetics elicits a female-specific burst of spontaneous movement, which requires ERα and Tac1 signaling. Disrupting the development of Nkx2-1(+) VMHVL neurons results in female-specific obesity, inactivity, and loss of VMHVL neurons coexpressing ERα and Tac1. Unexpectedly, two responses controlled by ERα(+) neurons, fertility and brown adipose tissue thermogenesis, are unaffected. We conclude that a dedicated subset of VMHVL neurons marked by ERα, NKX2-1, and Tac1 regulates estrogen-dependent fluctuations in physical activity and constitutes one of several neuroendocrine modules that drive sex-specific responses."

(Zapata et al., 2022)

Epigenetic modifier Kdm6a/Utx controls the specification of hypothalamic neuronal subtypes in a sex-dependent manner

"Ngn3 is a developmental regulator of energy homeostasis systems in the hypothalamus, pancreas, and intestine, whereas Pomc+ and Npy+ hypothalamic neurons shape a central microcircuitry for the integration of peripheral signals providing feedback on energy status and for the control of feeding and energy expenditure"

(Street et al., 2018)

Current Knowledge on Endocrine Disrupting Chemicals (EDCs) from Animal Biology to Humans, from Pregnancy to Adulthood: Highlights from a National Italian Meeting

> *"The MSH system, is sexually dimorphic with females having increased responsiveness to leptin and decreased responsiveness to insulin in comparison to males (Shi et al., 2008). The NPY/AgRP circuit is also sexually dimorphic. In particular, NPY immunoreactivity is sexually dimorphic in the ARC, the dorsomedial hypothalamus, and the PVN (Bo et al., 2016) and the Neuropeptide Y receptor Y1 (NPY-Y1) expression is higher in females compared to males (Martini et al., 2011)."*